2023BCSE008

Lab 04 Date:- 26|03|2026

Q1. Write a Java program to accept an integer value from the user and convert it into a byte. Display the result and observe the effect when the entered value exceeds the byte range.

Lab_04 > IntToByte.java > IntToByte

```java
//1. Write a Java program to accept an integer value from the user and convert it into a byte.
// Display the result and observe the effect when the entered value exceeds the byte range.

package Lab_04;
import java.util.*;

public class IntToByte {
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print(s: "Enter an integer: ");
        int num = sc.nextInt();
        byte b = (byte) num;
        System.out.println("Integer value: " + num);
        System.out.println("Converted byte value: " + b);
        sc.close();
    }
}
```

OUTPUT:—-

```
gauravmathur@Annus-Mac Lab_04 % java IntToByte.java
Enter an integer: 22
Integer value: 22
Converted byte value: 22
gauravmathur@Annus-Mac Lab_04 %
```

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Q2. Write a Java program to create a class Box with:

- Instance variables: length, breadth, height
- Constructors:
  - Default constructor
  - Parameterized constructor
- Method: volume()

Create objects using both constructors and display the volume.

Lab_04 > Box.java > Box

```java
/*2. Write a Java program to create a class Box with:
• Instance variables: length, breadth, height
• Constructors:
o Default constructor
o Parameterized constructor
• Method: volume()
Create objects using both constructors and display the volume. */
package Lab_04;
class Box {
    int length, breadth, height;
    Box() {
        length = 1;
        breadth = 1;
        height = 1;
    }
    Box(int l, int b, int h) {
        length = l;
        breadth = b;
        height = h;
    }

    int volume() {
        return length * breadth * height;
    }

    Run | Debug
    public static void main(String[] args) {
        Box b1 = new Box();
        Box b2 = new Box(l: 2, b: 3, h: 4);

        System.out.println("Volume (default): " + b1.volume());
        System.out.println("Volume (parameterized): " + b2.volume());
    }
}
```

OUTPUT:-

```
gauravmathur@Annus-Mac Lab_04 % JAVA Box.java
Volume (default): 1
Volume (parameterized): 24
gauravmathur@Annus-Mac Lab_04 %
```

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Q3. Write a Java program to create a class Dog with:

- Instance variables: name, age
- Method: display() to print details
- Initialize the object using:
  - Reference variable
  - Method
  - Constructor

Display the values for each initialization technique.

Lab_04 > Dog.java > Dog

```java
package Lab_04;
class Dog {
    String name;
    int age;
    void display() {
        System.out.println("Name: " + name + ", Age: " + age);
    }
    void setData(String n, int a) {
        name = n;
        age = a;
    }
    Dog(String n, int a) {
        name = n;
        age = a;
    }

    Dog() {}

    Run | Debug
    public static void main(String[] args) {

        // Reference var
        Dog d1 = new Dog();
        d1.name = "Tommy";
        d1.age = 3;
        d1.display();

        // Method
        Dog d2 = new Dog();
        d2.setData(n: "Rocky", a: 4);
        d2.display();

        // Constructor
        Dog d3 = new Dog(n: "Buddy", a: 5);
        d3.display();
    }
}
```

OUTPUT:-

```
gauravmathur@Annus-Mac LabJava %  /usr/bin/env /Library/Java/JavaVirtualMachines/jdk-19.
onMessages -cp /Users/gauravmathur/Library/Application\ Support/Code/User/workspaceStora
abJava_69e24954/bin Lab_04.Dog
Name: Tommy, Age: 3
Name: Rocky, Age: 4
Name: Buddy, Age: 5
gauravmathur@Annus-Mac LabJava %
```

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Q4. Write a Java program to accept a double value from the user and convert it into an integer using explicit type casting. Display both values.

Lab_04 > DoubleToInt.java > DoubleToInt

```java
//4. Write a Java program to accept a double value from the user and convert it into
//  Display both values.
package Lab_04;
import java.util.*;

public class DoubleToInt {
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print(s: "Enter a double value: ");
        double d = sc.nextDouble();
        int i = (int) d;
        System.out.println("Double value: " + d);
        System.out.println("Integer value: " + i);
        sc.close();
    }
}
```

OUTPUT:-

```
gauravmathur@Annus-Mac Lab_04 % JAVA DoubleToInt.java
Enter a double value: 22667888.09
Double value: 2.266788809E7
Integer value: 22667888
gauravmathur@Annus-Mac Lab_04 %
```

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Q5. Write a Java program to accept two numbers from the user: one integer and one double. Perform addition and display the result after implicit type conversion.

Lab_04 > ImplicitConversion.java > ImplicitConversion

```java
/*5. Write a Java program to accept two numbers from the user:
one integer and one double. Perform addition and display the result after implicit type conversion.
 */
package Lab_04;
import java.util.*;
public class ImplicitConversion {
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print(s: "Enter integer: ");
        int a = sc.nextInt();
        System.out.print(s: "Enter double: ");
        double b = sc.nextDouble();
        double result = a + b;
        System.out.println("Result: " + result);
        sc.close();
    }
}
```

OUTPUT:-

```
gauravmathur@Annus-Mac Lab_04 % JAVA ImplicitConversion.java
Enter integer: 23
Enter double: 2988.06
Result: 3011.06
gauravmathur@Annus-Mac Lab_04 %
```

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Q6. Write a Java program to demonstrate type promotion by multiplying two byte variables and storing the result in an integer variable. Display the result.

Lab_04 > TypePromotion.java > ...

```java
//6. Write a Java program to demonstrate type promotion by mu
package Lab_04;
import java.util.Scanner;
public class TypePromotion {
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println(x: "Enter the first number:");
        int a = sc.nextByte();
        System.out.println(x: "Enter second number:");
        int b = sc.nextByte();
        int result = a * b; // promoted to int
        System.out.println("Result: " + result);
        sc.close();
    }
}
```

OUTPUT:-

```
gauravmathur@Annus-Mac Lab_04 % JAVA TypePromotion.java
Enter the first number:
22
Enter second number:
34
Result: 748
gauravmathur@Annus-Mac Lab_04 %
```

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Q7. Write a Java program to accept a character from the user, convert it into its corresponding integer value, and display the result.

LabJava / Lab_04 / CharToInt.java / CharToInt

```java
//7. Write a Java program to accept a character from the use

package Lab_04;
import java.util.*;
public class CharToInt {
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print(s: "Enter a character: ");
        char ch = sc.next().charAt(index: 0);
        int value = ch;
        System.out.println("Character: " + ch);
        System.out.println("Integer value: " + value);
        sc.close();
    }
}
```

OUTPUT:-

```
gauravmathur@Annus-Mac Lab_04 % java CharToInt.java
Enter a character: A
Character: A
Integer value: 65
gauravmathur@Annus-Mac Lab_04 %
```

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Q8. Write a Java program to create a class Student with:

• Instance variables: name, marks

• Static variable: collegeName

• Methods:

- input()
- display()

Create **two objects** and demonstrate that the static variable is shared among all objects.

.abJava > Lab_04 > Student.java > Student > main(String[])

```java
/*8. Write a Java program to create a class Student with:
• Instance variables: name, marks
• Static variable: collegeName
• Methods:
o input()
o display() */
package Lab_04;
import java.util.*;
public class Student {
    String name;
    int marks;
    static String collegeName = "ABC College";

    void input(String n, int m) {
        name = n;
        marks = m;
    }

    void display() {
        System.out.println("Name: " + name);
        System.out.println("Marks: " + marks);
        System.out.println("College: " + collegeName);
    }

    Run | Debug
    public static void main(String[] args) {
        Student s1 = new Student();
        Student s2 = new Student();

        s1.input(n: "Annu", m: 90);
        s2.input(n: "Rahul", m: 85);

        // change static variable
        Student.collegeName = "NIT Srinagar";

        s1.display();
        System.out.println();
        s2.display();
    }
}
```

OUTPUT:-

```
gauravmathur@Annus-Mac Lab_04 % java Student.java
Name: Annu
Marks: 90
College: NIT Srinagar

Name: Rahul
Marks: 85
College: NIT Srinagar
gauravmathur@Annus-Mac Lab_04 %
```

Update is ready, click to restart